

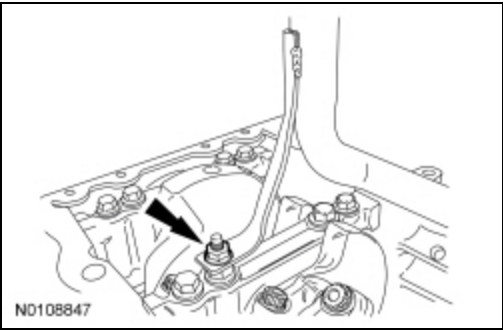
Oil Pump



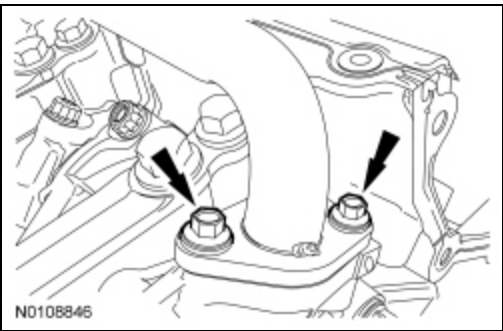
Material	
Item	Specification
Motorcraft® Metal Surface Prep Wipes ZC-31-B	—
Motorcraft® SAE 5W-20 Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-20 Super Premium Motor Oil (Canada) XO-5W20-QSP (US); CXO-5W20-LSP12 (Canada)	WSS-M2C945-A

Removal

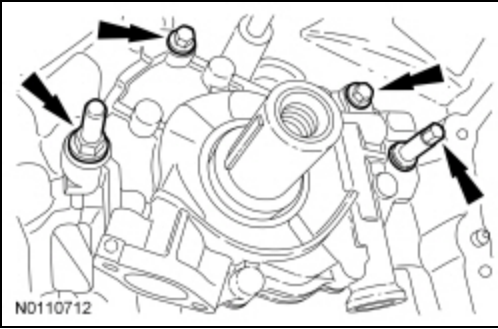
1. Remove the timing drive components. For additional information, refer to Timing Drive Components in this section.
2. Remove the oil pan. For additional information, refer to Oil Pan in this section.
3. Remove the oil pump screen and pickup tube support bracket nut.



4. Remove the 2 bolts and the oil pump screen and pickup tube.



5. Remove the 2 upper bolts, the 2 lower stud bolts and the oil pump.

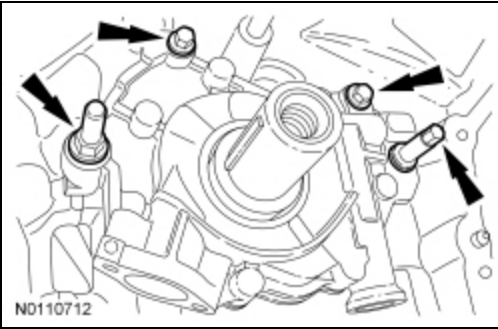


Installation

1. Prime the oil pump. Add 2 tablespoons of clean engine oil to the oil pump and rotate the oil pump by hand.
2. **NOTICE: Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean the sealing surfaces. These tools cause scratches and gouges which make leak paths. Use a plastic scraping tool to remove all traces of old sealant.**

Clean the sealing surfaces with metal surface prep. Follow the directions on the packaging. Inspect the mating surfaces.

3. Position the oil pump and install the 2 upper bolts and the 2 lower stud bolts in 5 stages.
 - Stage 1: Tighten all fasteners to 2 Nm (18 lb-in).
 - Stage 2: Tighten the 2 upper bolts to 10 Nm (89 lb-in).
 - Stage 3: Tighten the 2 lower stud bolts to 20 Nm (177 lb-in).
 - Stage 4: Tighten the 2 upper bolts an additional 45 degrees.
 - Stage 5: Tighten the 2 lower stud bolts an additional 60 degrees.

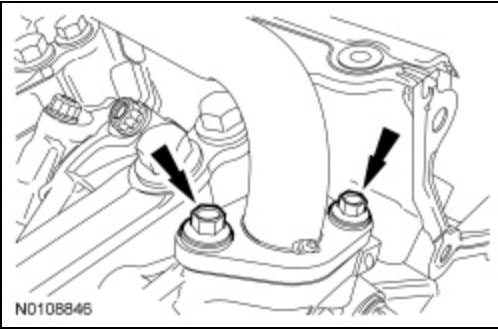


4. **NOTICE: Make sure the O-ring is in place and not damaged. A missing or damaged O-ring can cause foam in the lubrication system, low oil pressure and severe engine damage.**

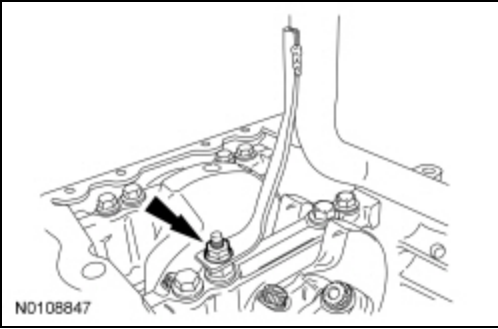
NOTE: Clean and inspect the mating surfaces and install a new O-ring. Lubricate the O-ring with clean engine oil prior to installation.

Position the oil pump screen and pickup tube and install the 2 front bolts.

- Tighten to 10 Nm (89 lb-in).



5. Install the oil pump screen and pickup tube support bracket nut.
 - Tighten to 24 Nm (18 lb-ft).



6. Install the oil pan. For additional information, refer to Oil Pan in this section.

7. Install the timing drive components. For additional information, refer to Timing Drive Components in this section.